MOBILE ROBOT KINEMATICS

WHAT ARE WE GOING TO LEARN?

- Introduction
- Degree of Freedom
- Differential Kinematics
  - Forward differential kinematics
  - Inverse differential kinematics

KINEMATICS

- Kinematics is a study of motion without considering the forces/efforts that affect the motion.

  - Deals with the geometric relationships that govern the system
  - Deals with the relationship between control parameters and the behaviour of a system in state space.

NEED OF MATHEMATICAL MODEL

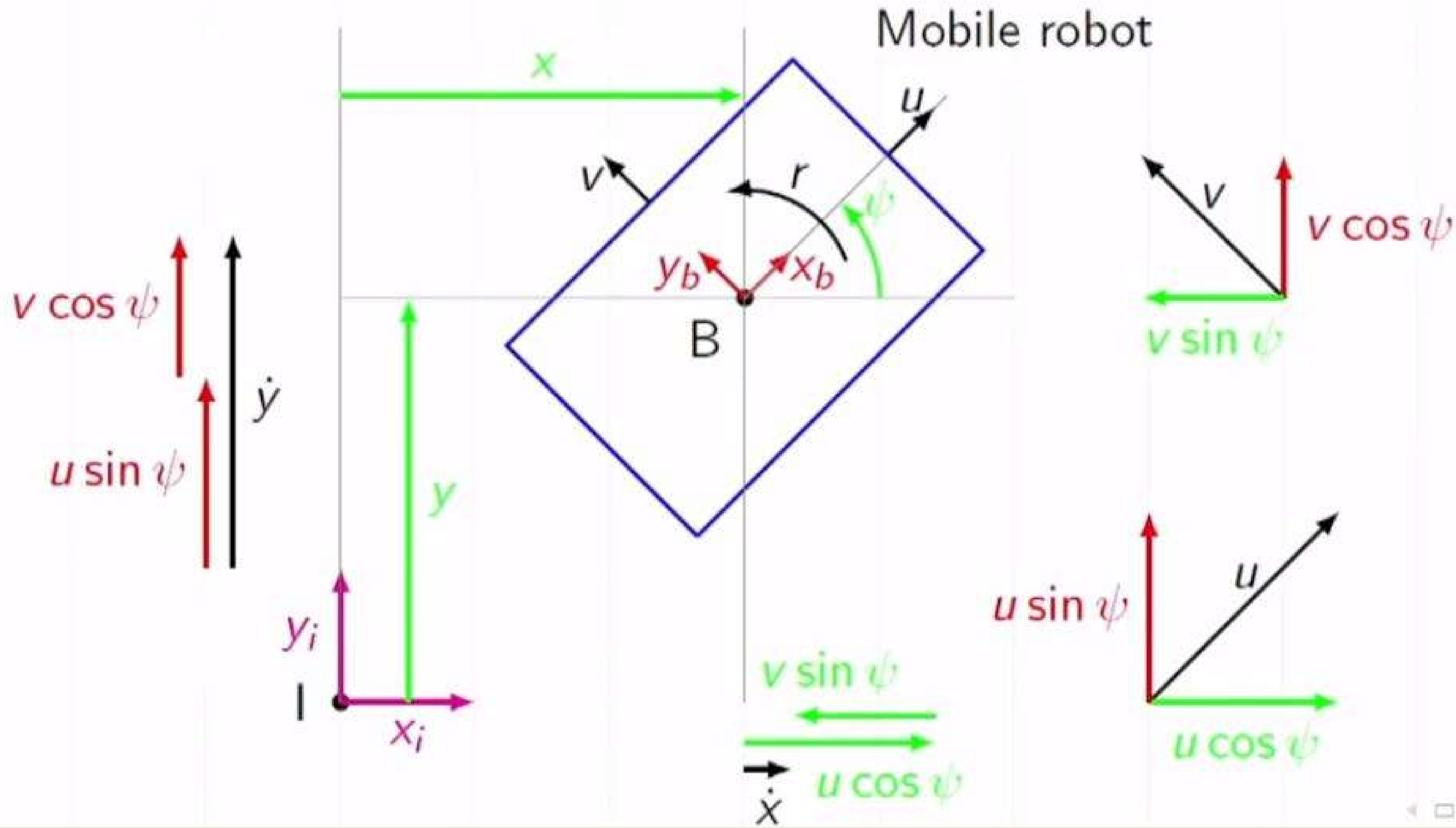
- To understand the behaviour of the system and design the system.
- To design suitable controllers, navigation systems and adjust their performances,
- To predict or estimate the system parameters, to illustrate or mimic or simulate the real system, etc.

DEGREE OF FREEDOM

- Minimum number of variables to describe the system.
  - For the land based wheeled mobile robots and water surface vehicles: Degree of freedom is three(two translations and one orientation in a plane)
  - For the aerial, space and underwater robots: degree of freedom is six(three translation and three orientation in space)

WHEELED MOBILE ROBOTS

- WMRs, as the name implies, have the ability to move around(on the ground surface) with the help of wheels.
- DoF is three for the ground/ land- based wheeled mobile robots. i.e. two translations and one orientation in a plane.


Mobile robot
x
u
v
r
ψ
y_b
x_b
B
v cos ψ
u sin ψ
ẏ
y
y_i
I
x_i
v sin ψ
u cos ψ
ẋ
v
v cos ψ
v sin ψ
u
u sin ψ
u cos ψ

x: Forward displacement of the mobile robot w.r.t. I

y: Lateral displacement of the mobile robot w.r.t. I

ψ: Angular displacement of the mobile robot w.r.t. I

u: Forward velocity of the mobile robot w.r.t. B

v: Lateral velocity of the mobile robot w.r.t. B

r: Angular velocity of the mobile robot w.r.t. B

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} u \cos\psi - v \sin\psi \\ u \sin\psi + v \cos\psi \\ r \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} u\cos\psi - v\sin\psi \\ u\sin\psi + v\cos\psi \\ r \end{bmatrix} = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix}$$

$$\dot{\boldsymbol{\eta}} = \mathbf{J}(\psi)\,\boldsymbol{\zeta}$$

It describes the relation between the velocity input commands (ζ) and the derivatives of generalized coordinates ($\dot{\eta}$).

FORWARD DIFFERTIAL KINEMATICS

- For given input velocity commands, finding the derivatives of generalized coordinates

$$\dot{\eta} = \mathbf{J}(\psi)\zeta$$

- Simulating or analysing the system in velocity level

INVERSE DIFFERENTIAL KINEMATICS

- For the desired derivatives of generalized coordinates (or given position trajectory), finding the corresponding velocity input commands.

$$\zeta = \mathbf{J}^{-1}(\psi)\,\dot{\eta}$$

- Controlling the system in velocity level